

Thermodynamic characterization of second generation biofuels: Vapour–liquid equilibria and excess enthalpies of the binary mixtures 1-pentanol and cyclohexane or toluene

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ABSTRACT

The paper reports accurate experimental data of vapour–liquid equilibria at $T = 313.15$ K, using an isothermal total pressure cell, for the binary systems (1-pentanol + cyclohexane) and (1-pentanol + toluene). The data were correlated using Margules, Wilson, NRTL and UNIQUAC equations. Also, the enthalpies of mixing were measured for the same systems at two different temperatures $T = (298.15$ and $313.15)$ K using an isothermal flow calorimeter and the excess enthalpies were correlated by the Redlich–Kister equation. The thermodynamic behaviour of both systems is discussed.

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1. Introduction

The European Union is strongly dependent on fossil fuels for its transport needs so that the EU has adopted measures to encourage the production and use of sustainable biofuels because they can provide a very significant contribution in the short to medium term. Furthermore, biofuels can contribute to the EU's objectives of securing the EU fuel supply while improving the greenhouse gas balance and fostering the development of a competitive European (biofuels and other) industry.

To facilitate the wider use of biofuels in road transport, biofuels and blends need to become increasingly compatible with existing logistics as well as future ones but a good characterization of the thermodynamic properties is needed to obtain an in-depth understanding for example of the relationship between biofuels quality and engine performance.

In this context, 1-pentanol is a very important compound in the second generation of biogasolines due to the fact that it has high octane rating, up to 100 research octane number (RON), better water tolerance and very high heat of combustion.

This paper reports accurate experimental data of vapour–liquid equilibria and enthalpies of mixing of the mixtures (1-pentanol + cyclohexane) and (1-pentanol + toluene) in order to contribute to a better knowledge of second generation biofuels.

2. Experimental

2.1. Materials

The compounds were purchased from Sigma–Aldrich and were of the highest purity available, chromatography quality reagents with a purity >0.99 (GC) for 1-pentanol, >0.995 (GC) for cyclohexane and >0.998 (GC) for toluene. Their purities were also checked by gas chromatography and all were found to be >0.997 . These details are summarized in Table 1.

2.2. Experimental techniques

A static VLE apparatus, consisting of an isothermal total pressure cell, has been employed for measuring the vapour–liquid equilibrium of the binary mixture. The apparatus and measuring technique has been described in a previous paper [1]. Experimental values of total vapour pressure were obtained in two overlapping runs starting from opposite ends of the composition range. The equilibrium properties were measured directly and their uncertainties are: injected volume ± 0.03 ml, temperature ± 10 mK and total pressure

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Table 1

Material description.

Chemical name	Source	Mass fraction purity	Purification method
1-Pentanol	Sigma-Aldrich	≥0.997	None
Cyclohexane	Sigma-Aldrich	≥0.997	None
Toluene	Sigma-Aldrich	≥0.998	None

±5 Pa. The overall mole fraction is determined accurately from the volumetric displacement of the injector pistons, the temperature of the injectors and the densities for the pure components. The estimated uncertainties for mole fractions are ±0.0005.

Excess enthalpies have been measured with a quasi-isothermal flow calorimeter, developed in our laboratory [2]. It is based on the accurate measurement of the energy required to maintain the mixing vessel at a constant temperature. To achieve this condition, a Peltier cooler removes, at constant rate, energy from the flow cell and a control-heater compensates this energy and additionally the energy liberated (exothermic mixing) or absorbed (endothermic mixing) by the mixing process. The properties measured directly and their corresponding uncertainties are: flow rate ±0.15% (1–10 mL/min); temperature ±10 mK; the heating and cooling systems have stability better than 2.4 μW, and it results in an uncertainty of ±0.0001 in the composition of the mixture and the extended uncertainty in the excess molar enthalpy is better than ±1% for a cover factor of 2 which means a confidence level of 0.95.

3. Results

The use of a static technique for measuring the equilibrium means that the vapour phase need not be sampled for analysis and the data are thermodynamically consistent “per se” [3]. Data reduction was done by Barker’s method according to well-established procedures [4,5]. The non-ideality of the vapour phase was taken into account with the virial equation of state, truncated after the second term. The pure component and interaction second virial coefficients (B_{ii} , B_{ij}) were calculated by the Hayden and O’Connell method [6] using the parameters given by Dymond and Smith [7].

VLE data were correlated by the Margules equation up to six parameter [8]:

$$\frac{G_m^E}{RT} = [A_{ij}x_i + A_{ij}x_j - (\lambda_{ji}x_i + \lambda_{ij}x_j)x_i x_j + (\eta_{ji}x_i + \eta_{ij}x_j)x_i^2 x_j^2]x_i x_j \quad (1)$$

Also Wilson [9], NRTL [10] and UNIQUAC [11] models were used for the correlation, which are given by Eqs. (2)–(4), respectively:

$$\frac{G_m^E}{RT} = -\sum_i x_i \ln \left(\sum_j x_j A_{ij} \right) \quad (2)$$

$$\frac{G_m^E}{RT} = \sum_i x_i \left(\frac{\sum_j A_{ji} G_{ji} x_j}{\sum_k G_{ki} x_k} \right) \quad (3)$$

$$\frac{G_m^E}{RT} = \sum_i x_i \ln \left(\frac{\varphi_i}{x_i} \right) + \left(\frac{z}{2} \right) \sum_i q_i x_i \ln \left(\frac{\vartheta_i}{q_i} \right) - \sum_i q_i x_i \ln \left(\sum_j \vartheta_j A_{ji} \right) \quad (4)$$

where $G_{ji} = \exp(-\alpha_{ji} A_{ji})$; $\alpha_{ji} = 0.3$; $\vartheta_i = (q_i x_i) / \sum_j q_j x_j$; $\varphi_i = (r_i x_i) / \sum_j r_j x_j$ and $z = 10$.

Table 2Total pressure VLE data for the binary system 1-pentanol (1) + cyclohexane (2) at 313.15 K^a.

x_1	$y_{1,calc}$	P (kPa)	x_1	$y_{1,calc}$	P (kPa)
0.0000	0.0000	24.569	0.5012	0.0290	20.606
0.0491	0.0143	24.329	0.5492	0.0311	19.826
0.1037	0.0172	24.071	0.5514	0.0312	19.931
0.1492	0.0183	23.788	0.5993	0.0339	18.880
0.1989	0.0195	23.485	0.6016	0.0341	18.856
0.2492	0.0210	23.139	0.6517	0.0378	17.716
0.2993	0.0225	22.764	0.7025	0.0428	16.322
0.3493	0.0240	22.330	0.7517	0.0495	14.712
0.3991	0.0255	21.846	0.8019	0.0594	12.754
0.4010	0.0256	21.832	0.8516	0.0753	10.462
0.4494	0.0272	21.276	0.9015	0.1063	7.746
0.4511	0.0272	21.267	0.9512	0.1918	4.560
0.4993	0.0290	20.614	1.0000	1.0000	0.900

^a Standard uncertainties ($k=1$): $u(x_1) = 0.0005$, $u(y_{1,calc}) = 0.0005$, $u(P) = 5$ Pa; $u(T) = 10$ mK.

Table 3Total pressure VLE data for the binary system 1-pentanol (1) + toluene (2) at 313.15 K^a.

x_1	$y_{1,calc}$	P (kPa)	x_1	$y_{1,calc}$	P (kPa)
0.0000	0.0000	7.864	0.5052	0.0904	6.516
0.0493	0.0283	7.789	0.5497	0.0972	6.268
0.0991	0.0428	7.698	0.5551	0.0981	6.269
0.1495	0.0513	7.595	0.5997	0.1062	5.968
0.1997	0.0571	7.485	0.6050	0.1073	5.966
0.2495	0.0620	7.365	0.6548	0.1184	5.638
0.2995	0.0667	7.232	0.7046	0.1326	5.230
0.3497	0.0716	7.082	0.7539	0.1514	4.760
0.3997	0.0770	6.918	0.8040	0.1788	4.203
0.4049	0.0776	6.922	0.8532	0.2211	3.555
0.4497	0.0829	6.723	0.9023	0.2953	2.799
0.4551	0.0836	6.725	0.9517	0.4571	1.917
0.4996	0.0896	6.508	1.0000	1.0000	0.901

^a Standard uncertainties ($k=1$): $u(x_1) = 0.0005$, $u(y_{1,calc}) = 0.0005$, $u(P) = 5$ Pa; $u(T) = 10$ mK.

Tables 2 and 3 give the experimental values of total pressure, the liquid phase composition and the calculated vapour phase composition by Margules equation for the binary systems (1-pentanol + cyclohexane) and (1-pentanol + toluene) respectively, at $T = 313.15$ K and both sets of data have been represented in Fig. 1. The average values of the experimental vapour pressures (P_i^{sat}) for the pure compounds and their comparison with the literature, the molar volumes of pure liquids (V_i^L) and the second virial coefficients (B_{ii} , B_{ij}) are indicated in Table 4. The results of VLE data correlation

Table 4Average values of the experimental vapour pressures (P_i^S) for the pure compounds and literature values ($P_i^S(\text{lit.})$), the molar volumes of pure liquids (V_i^L) and the second virial coefficients (B_{ii} , B_{ij}) at $T = 313.15$ K used for the calculations.

	1-Pentanol ($i=1$)	Cyclohexane ($i=2$)	Toluene ($i=3$)
P_i^S (kPa)	0.908	24.630	7.897
P_i^S (lit.) (kPa)	0.918 ^a 0.872 ^b 0.891 ^c	24.630 ^d 24.635 ^e	7.880 ^a 7.894 ^e 7.861 ^f
V_i^L (cm ³ mol ⁻¹) ^g	110	111	109
B_{ii} (cm ³ mol ⁻¹) ^h	−3001	−1925	−2043
B_{ij} (cm ³ mol ⁻¹) ^h	−3001	−1508	−2250

^a Ref. [20].

^b Ref. [21].

^c Ref. [22].

^d Ref. [23].

^e Calculated from Antoine equation using constants reported in Ref. [24].

^f Calculated from Antoine equation using constants reported in Ref. [8].

^g Ref. [25].

^h Calculated by Hayden et al. [6] from Dymond et al. [7].

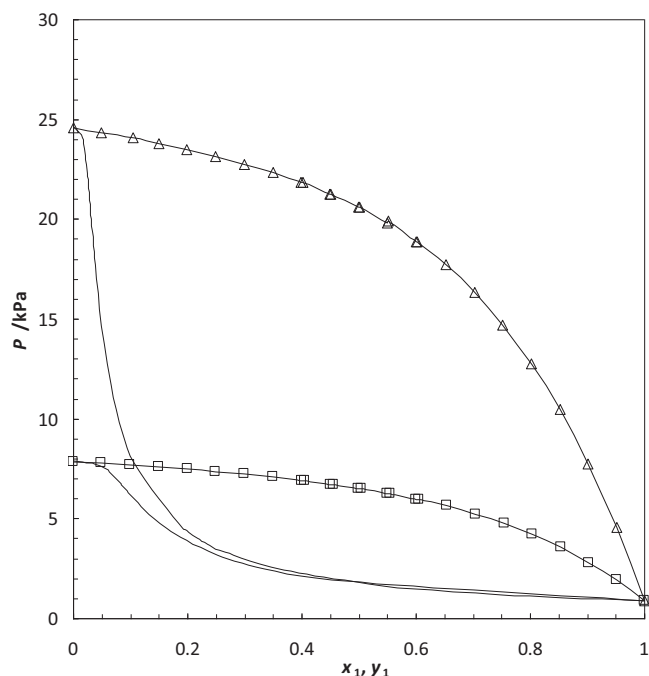


Fig. 1. Experimental VLE data at 313.15 K for the systems 1-pentanol (1)+cyclohexane (2): (Δ) and 1-pentanol (1)+toluene (2): ($\square$). Calculated values using Margules equation with the parameters given in Table 5: (—).

are summarized in Table 5; it contains the values of the adjustable parameters for the different models, the root mean square of the difference between the experimental and the calculated pressures (rms ΔP) and the maximum value of this difference (max $|\Delta P|$).

For the same two binary mixtures the excess enthalpies were measured at $T=(298.15 \text{ and } 313.15) \text{ K}$, these experimental data as a function of the composition are given in Table 6 and Fig. 2. The data have been correlated using Redlich–Kister equation:

$$\frac{H_m^E}{\text{J mol}^{-1}} = x(1-x) \sum_{i=1}^n \frac{A_i(2x-1)^{i-1}}{1+C(2x-1)} \quad (5)$$

Table 5

Determined parameters of the models used for the correlation of the binary systems 1-pentanol (1)+cyclohexane (2) and 1-pentanol (1)+toluene (2) at $T=313.15 \text{ K}$, together with the root mean square deviation of pressure (rms ΔP) and the maximum value of the deviation (max $|\Delta P|$). The ΔP term is defined as the difference between the experimental and calculated pressure.

	Margules	Wilson	NRTL	UNIQUAC
1-Pentanol (1) + cyclohexane (2)				
A_{12}	2.8605	0.0765	0.6461	1.2705
A_{21}	1.1470	0.7443	2.1906	0.4583
λ_{12}	5.7913			
λ_{21}	0.4046			
η_{12}	8.7510			
η_{21}	0.2810			
α_{12}			0.5630	
rms ΔP (kPa)	0.054	0.088	0.137	0.313
max $ \Delta P $ (kPa)	0.114	0.228	0.330	0.650
1-Pentanol (1) + toluene (2)				
A_{12}	1.9594	0.1705	0.5156	1.1491
A_{21}	1.0374	0.7702	1.5908	0.6050
λ_{12}	1.2906			
λ_{21}	0.2011			
α_{12}			0.6024	
rms ΔP (kPa)	0.018	0.018	0.019	0.047
max $ \Delta P $ (kPa)	0.033	0.034	0.033	0.086

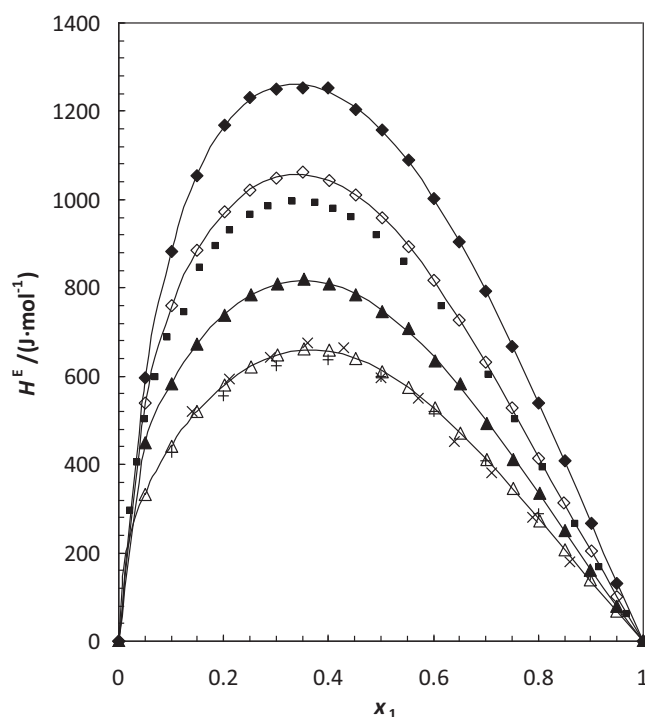


Fig. 2. Experimental molar excess molar enthalpies as a function of the composition for the binary systems: 1-pentanol (1)+toluene (2) ($\diamond$) at $T=298.15 \text{ K}$ and ($\blacklozenge$) at $T=313.15 \text{ K}$; 1-pentanol (1)+cyclohexane (2) (Δ) at $T=298.15 \text{ K}$ and ($\blacktriangle$) at $T=313.15 \text{ K}$; and literature values 1-pentanol (1)+toluene (2) ($\blacksquare$) at $T=298.15 \text{ K}$ (Ref. [16]) and 1-pentanol (1)+cyclohexane (2) at $T=298.15 \text{ K}$ ($\times$) Ref. [17]; ($+$) Ref. [18].

The parameters of the equation and the standard deviations are summarized in Table 7, the optimal number of parameters was selected by examining F -test.

4. Discussion

The experimental data on vapour–liquid equilibria have been well correlated with all the models. For the binary 1-pentanol + cyclohexane the six-parameter Margules equation gives the best results with a root mean square pressure residual of 54 Pa and a maximum deviation of 114 Pa. But for the system 1-pentanol + toluene, the four-parameter Margules equation, and the models of Wilson and NRTL obtain quite similar results. For both systems the UNIQUAC model gives the worst fit.

In Fig. 1, the total pressure is plotted as a function of the molar fraction of the liquid and the vapour phases; it is shown that both systems exhibit a positive deviation from the ideality and the effect is higher for the mixture with cyclohexane, but neither of them presents an azeotrope. The addition of the alcohol to the hydrocarbons produces the decrease of the vapour pressure.

There are some VLE data in the literature for the binary mixtures 1-pentanol + cyclohexane [12,13] and 1-pentanol + toluene [14,15], but they were measured at different conditions and they cannot be compared.

Concerning the effect of mixing, both systems have a strong endothermic behaviour which increases at higher temperatures and the highest excess enthalpy is observed for the system 1-pentanol + toluene at 313.15 K. The maximum value is $H_m^E = 1252 \text{ J mol}^{-1}$ for a mole fraction of 1-pentanol of 0.35 at $T=313.15 \text{ K}$ and it decreases to $H_m^E = 1062 \text{ J mol}^{-1}$ at the same composition at $T=298.15 \text{ K}$. On the other hand, the binary mixture 1-pentanol + cyclohexane shows the maximum excess enthalpy at a mole fraction of 1-pentanol of 0.35 (the same that the other binary

Table 6Experimental excess molar enthalpies H_m^E as a function of the mole fraction x_1 for the binary systems^a.

x_1	H_m^E (J mol ⁻¹)	x_1	H_m^E (J mol ⁻¹)	x_1	H_m^E (J mol ⁻¹)
1-Pentanol (1) + cyclohexane (2) at $T = 298.15$ K					
0.0000	0.0	0.3523	660.9	0.7022	411.0
0.0500	333.3	0.4023	658.4	0.7522	347.2
0.1001	440.1	0.4523	641.4	0.8021	271.9
0.1501	520.5	0.5022	609.2	0.8501	206.1
0.2022	581.0	0.5522	573.5	0.9000	138.1
0.2522	622.0	0.6022	527.4	0.9500	67.7
0.3022	648.7	0.6522	471.1	1.0000	0.0
1-Pentanol (1) + cyclohexane (2) at $T = 313.15$ K					
0.0000	0.0	0.3521	818.7	0.7019	494.1
0.0500	449.0	0.4022	809.9	0.7521	412.1
0.1000	581.9	0.4522	783.5	0.8021	334.5
0.1501	673.3	0.5022	745.9	0.8501	250.2
0.2021	739.3	0.5522	709.5	0.9000	159.6
0.2522	785.3	0.6022	634.2	0.9500	79.8
0.3022	809.4	0.6519	583.1	1.0000	0.0
1-Pentanol (1) + toluene (2) at $T = 298.15$ K					
0.0000	0.0	0.3500	1062.7	0.7003	632.4
0.0511	539.3	0.4018	1042.3	0.7507	528.9
0.1004	760.3	0.4516	1010.7	0.8012	414.9
0.1497	883.9	0.5016	959.5	0.8497	312.5
0.2012	972.6	0.5517	893.7	0.9004	203.1
0.2507	1021.2	0.6018	816.0	0.9491	101.5
0.3003	1048.1	0.6500	728.1	1.0000	0.0
1-Pentanol (1) + toluene (2) at $T = 313.15$ K					
0.0000	0.0	0.3500	1252.4	0.7004	793.7
0.0511	596.4	0.3999	1253.6	0.7508	666.9
0.1004	883.7	0.4518	1202.6	0.8013	540.2
0.1498	1053.3	0.5016	1157.0	0.8498	409.2
0.2013	1168.8	0.5517	1089.3	0.9005	267.0
0.2508	1231.4	0.6019	1002.7	0.9492	132.0
0.3003	1251.1	0.6501	904.2	1.0000	0.0

^a Standard uncertainties ($k = 1$): $u(x_1) = 0.0001$, $u(T) = 10$ mK, $u(H_m^E) = 0.5\%$.

system) and the corresponding values are 819 J mol^{-1} at 313.15 K and 661 J mol^{-1} at 298.15 K .

We have found excess enthalpy data for both systems at 298.15 K in the literature. The system 1-pentanol+toluene was measured by Mrazek and Van Ness. [16]. The reduction of the data according to our procedure gives a root mean square deviation of 7.1 J mol^{-1} with a maximum deviation of 25.4 J mol^{-1} (removing the point of the highest deviation the values change to 0.9 and 1.6 J mol^{-1}). It can be observed in Fig. 2, that the literature values give a lower value for the excess enthalpies, the differences in the maximum effect are -68 J mol^{-1} which represents 6.8% and it does not agree with the uncertainty in the measurement.

The excess enthalpy of the binary 1-pentanol+cyclohexane at 298.15 K was also measured by González Posa et al. [17] and Miranda et al. [18]. We have correlated their data using our procedure and the results of the correlation give a root mean square deviation of 9.6 J mol^{-1} with a maximum deviation of 13.8 J mol^{-1} for the first data [17] and 2.8 J mol^{-1} and 4.9 J mol^{-1} , respectively for the second reference [18]. Our measurements improve the data and the number of experimental points. The difference between both sets of data at the maximum excess enthalpy ranges from

17 J mol^{-1} to 25 J mol^{-1} respectively. The agreement is good taking into account the uncertainties of both experimental equipments. Also some data at 313.15 K were found in the literature [19] but only at infinite dilution.

Finally the excess functions of the both binary systems at 313.15 K are represented in Figs. 3 and 4 for the binary systems 1-pentanol+cyclohexane and 1-pentanol+toluene, respectively. The excess Gibbs energies were calculated using the Margules equation and the excess enthalpies using the Redlich–Kister equation, and the excess entropies using the relation between them: $G^E = H^E - T \times S^E$. These functions can exhibit quite different behaviours and values and could be interpreted in terms of intermolecular forces. The presence of 1-pentanol with a high self-association implies a high non-ideal behaviour as can be seen in the figures. In both mixtures the attractive forces between different molecules are weaker than between the same molecules and it results in a high endothermic effect of mixing. This effect is also greater at higher temperatures. The binary 1-pentanol+cyclohexane, presents negative excess entropy in a wide range of composition which means a greater positive value of the excess Gibbs energy.

Table 7Parameters of Redlich–Kister equation, standard deviation of excess molar enthalpies σ and, absolute maximum deviation ($|\Delta H_{m,\max}^E|$) for the measured systems.

T (K)	A_1	A_2	A_3	A_4	A_5	C	σ	$ \Delta H_{m,\max}^E $
1-Pentanol (1) + cyclohexane (2) (J mol ⁻¹)								
298.15	2454.3	1115.8	-921.53	-96.311	0	0.98862	1.8	3.5
313.15	3006.5	1242.4	-1081.9	-197.16	0	0.97525	4.3	9.2
1-Pentanol (1) + toluene (2) (J mol ⁻¹)								
298.15	3849.2	1096.7	-1283.5	-345.68	350.87	0.88289	2.2	5.3
313.15	4632.7	1284.8	-667.20	-643.52	0	0.80392	4.5	8.3

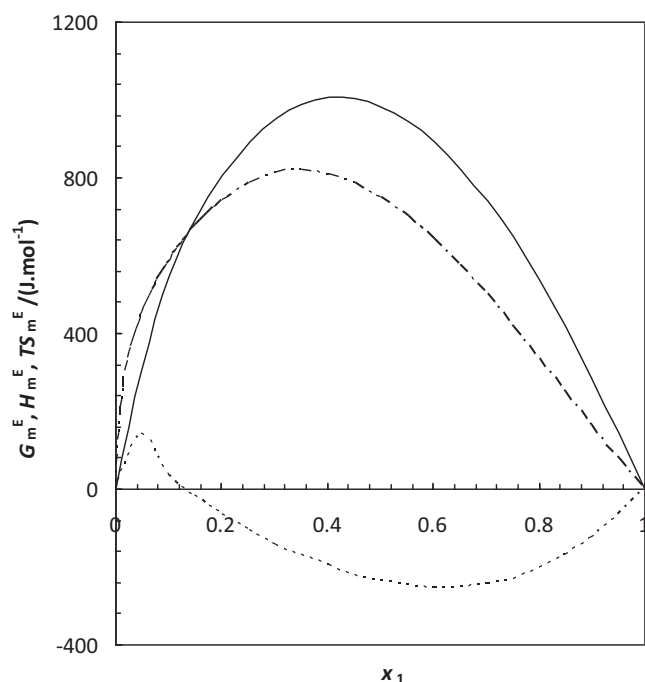


Fig. 3. Excess functions for the system 1-pentanol + cyclohexane at $T = 313.15$ K: (—) G_m^E ; (---) H_m^E ; (·····) TS_m^E .

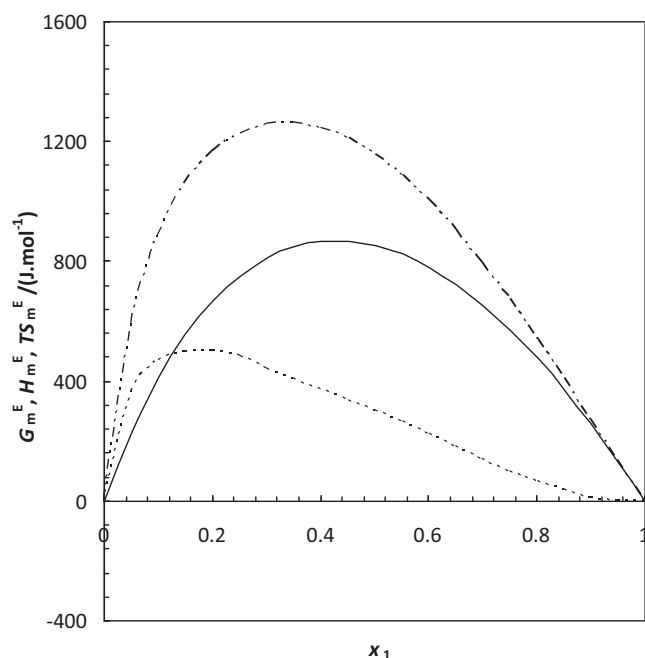


Fig. 4. Excess functions for the system 1-pentanol + toluene at $T = 313.15$ K: (—) G_m^E ; (---) H_m^E ; (·····) TS_m^E .

List of symbols

- A_i adjustable parameters of Redlich–Kister equation, Eq. (5)
 A_{ij}, A_{ji} adjustable parameters of the VLE correlation models, Eqs. (2–4)
 B_{ii}, B_{ij}, B_{jj} second virial coefficients

- C adjustable parameter of Redlich–Kister equation, Eq. (5)
 G_m^E excess molar Gibbs energy
 H_m^E excess molar enthalpy
 i, j constituent identification: 1 or 2
 lit. literature value
 max maximum value of the indicated quantity
 P total pressure
 P_i^s vapour pressure of pure constituent i
 R universal gas constant
 rms root mean square
 S_m^E excess molar entropy
 T absolute temperature
 V_i^L molar volume of pure liquid $i = 1, 2$
 x mole fraction, liquid phase
 y mole fraction, vapour phase

Greek letters

- Δ signifies difference
 α_{ij} adjustable parameter in NRTL model
 $\lambda_{ij}, \lambda_{ji}$ adjustable parameters in Eq. (1)
 η_{ij}, η_{ji} adjustable parameters in Eq. (1)

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